



Faculty of Computer Science

Data Science

Master's Thesis Defence 2026

Enhancing In-Context Time-Series Forecasting Through Feature Reconstruction and Transformer Token Optimization

Adeel Ahmed

Master's Program: Data Science
Faculty of Computer Science
HSE University

Supervisor: **Prof. Sabarathinam Srinivasan**

Assistant Professor
Faculty of Computer Science
HSE University

Motivation



Energy Forecasting

Demand prediction and grid optimization



Weather Prediction

Accurate forecasts for planning and safety



Financial Analysis

Market forecasting and risk management



Industrial Monitoring

Predictive maintenance and process control

Time-series forecasting is critical for data-driven decision making.

Challenges:

- Real-world time series contain noise and irregular fluctuations.
- Forecasting performance depends on the quality of temporal representations.
- Modern forecasting models often focus on architecture improvements.
- The impact of signal reconstruction before tokenization remains underexplored.

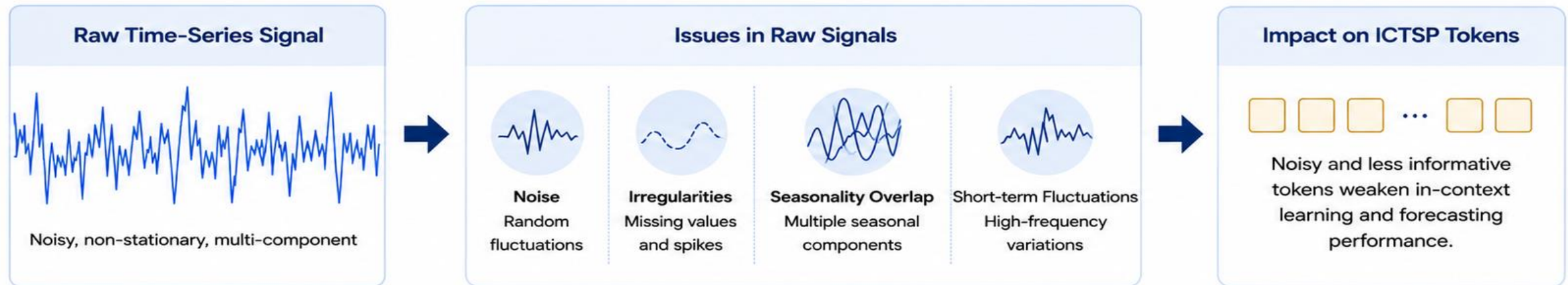


Key Question:

Can feature reconstruction improve the quality of temporal representations and enhance ICTSP forecasting performance?

Problem Statement

In-Context Time-Series Predictor (ICTSP) tokenizes raw time-series segments without preprocessing, which may contain noise, irregularities, seasonality overlap, and short-term fluctuations.



Research Problem:

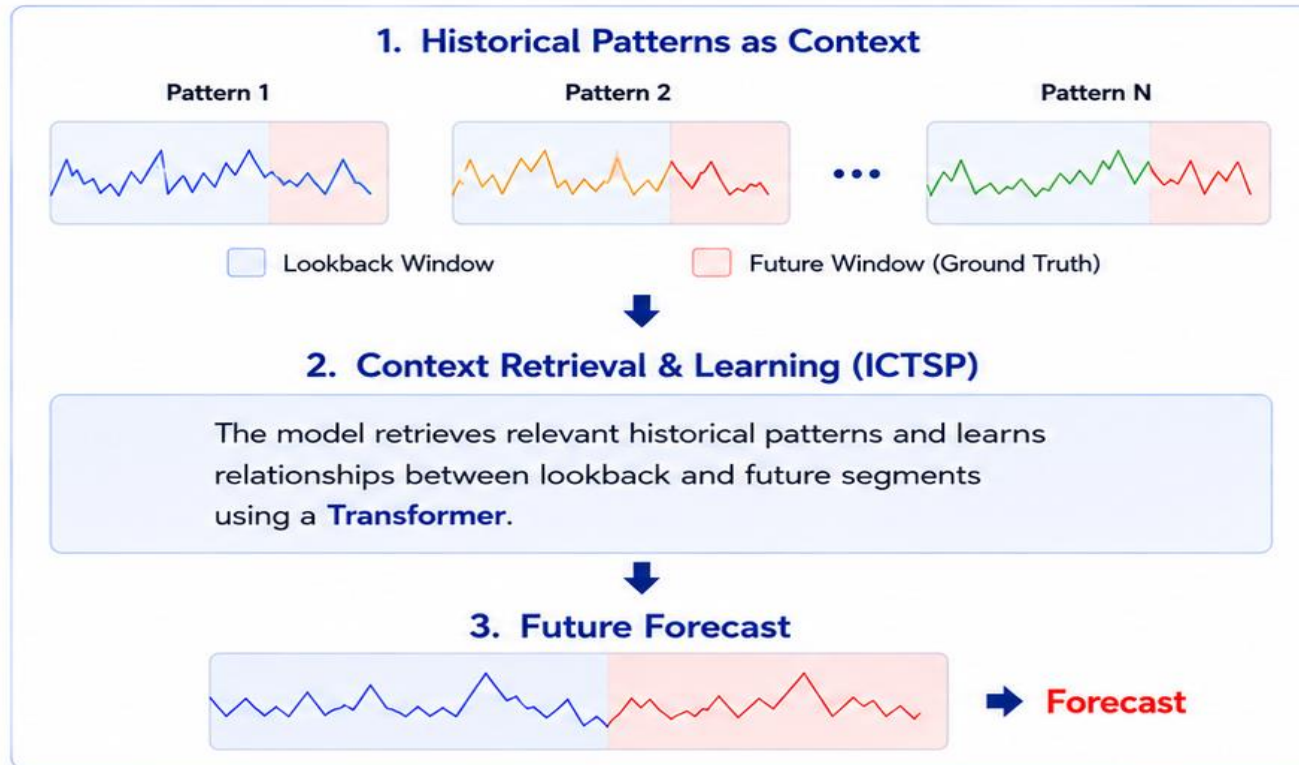
Can feature reconstruction techniques applied before ICTSP tokenization improve the quality and structure of forecasting-task tokens, and thus enhance ICTSP forecasting performance?

Research Gap:

Current literature has not explored whether pre-tokenization decomposition or reconstruction can enhance ICTSP performance.

In-Context Time Series Predictor (ICTSP) - Overview

ICTSP is a **foundation model** that reformulates time-series forecasting as an **in-context learning** problem by leveraging historical forecasting patterns.



Key Characteristics of ICTSP

- Treats time-series forecasting as an **in-context learning** task.
- Uses **(lookback, future)** pairs from historical data as **context examples**.
- Transformer model learns relationships between patterns and **adapts at inference time** without parameter updates.
- Generates future values by leveraging the most **relevant contextual patterns**.

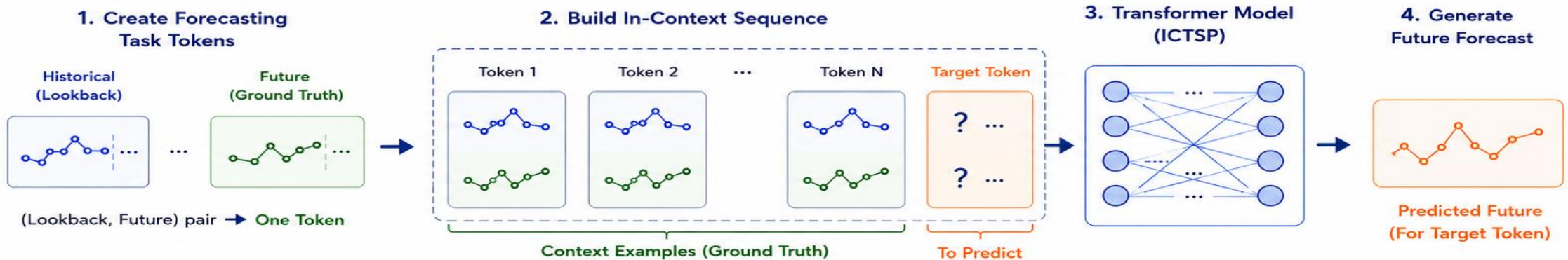
Key Insight

ICTSP performance strongly depends on the **quality of the generated forecasting-task tokens**. Better tokens lead to better context understanding and more accurate forecasts.

How ICTSP Works and Its Limitation

ICTSP treats forecasting as an **in-context learning** problem by using historical (**lookback, future**) pairs as tokens.

A. HOW ICTSP WORKS



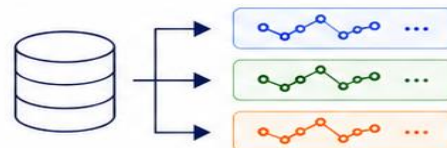
B. LIMITATIONS OF ICTSP

1. Sensitive to Input Quality



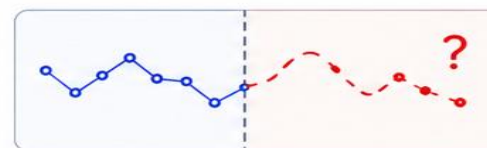
Noisy raw data create poor-quality tokens and reduce forecasting accuracy.

2. Depends on Similar Contexts



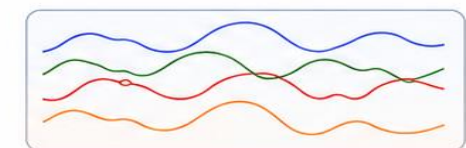
If similar historical patterns are unavailable, forecasting performance drops.

3. Limited Long-Term Generalization



Performance decreases when future patterns differ significantly from historical ones.

4. Multivariate Complexity



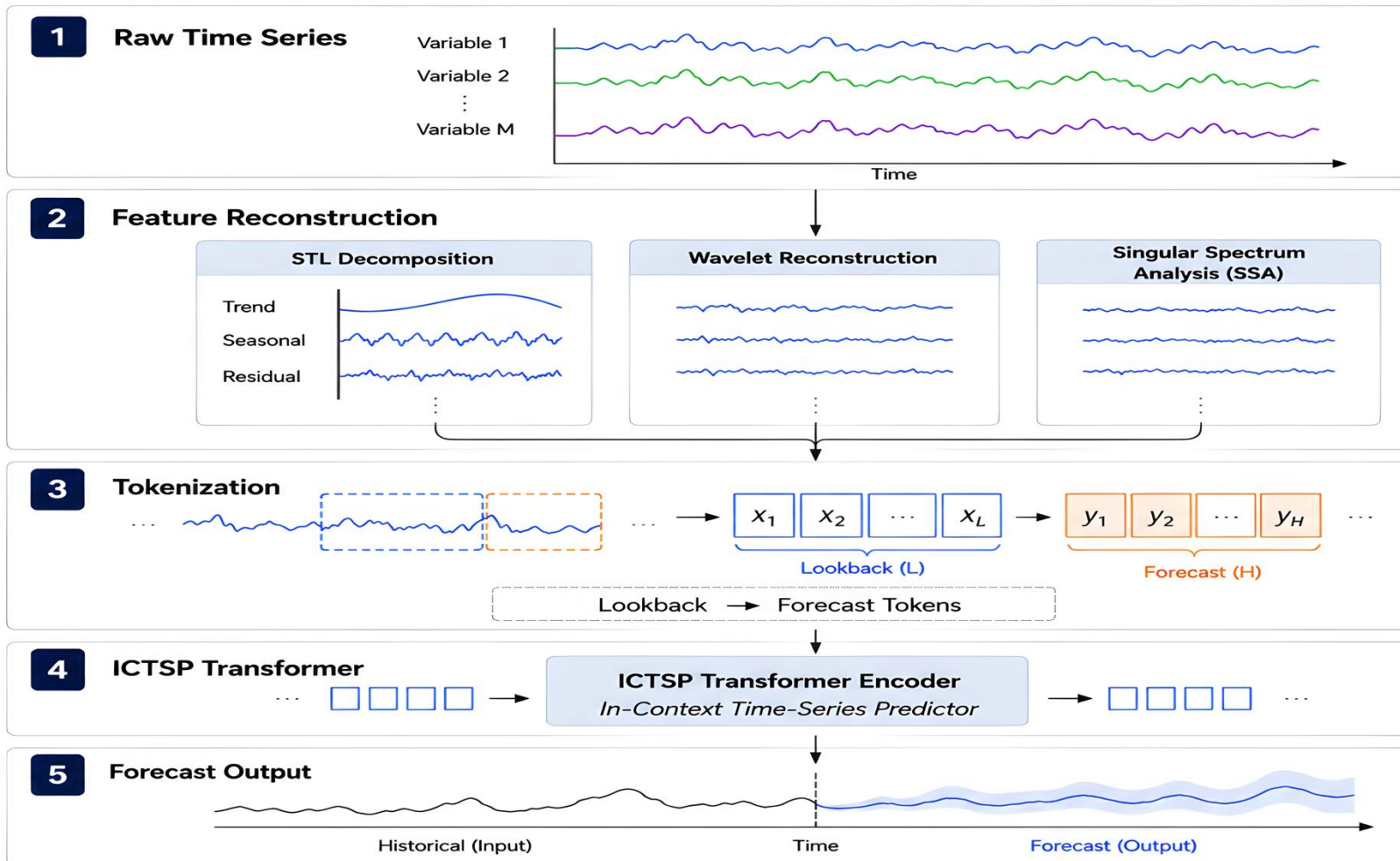
Complex dependencies between multiple variables make forecasting more difficult.

Key Takeaway:

ICTSP is a powerful zero-shot forecaster, but its performance depends heavily on **input token quality** and **relevant historical patterns** in context.

Proposed Feature Reconstruction Framework

We enhance ICTSP forecasting by reconstructing the input signal before tokenization while ICTSP architecture unchanged



Key Idea

- Improve signal quality before tokenization.
- Preserve important temporal patterns.
- Reduce noise and irregular fluctuations.
- Generate more informative forecasting tokens.

Important Design Choice

The original ICTSP architecture, training settings, and forecasting mechanism remain **unchanged**.

Research Objective

Evaluate whether feature reconstruction before tokenization improves ICTSP forecasting performance under full-data, few-shot, and zero-shot settings.

Feature Reconstruction Methods

Three complementary reconstruction techniques were selected to enhance different characteristics of temporal signals before **ICTSP tokenization**.

	STL (Seasonal-Trend Decomposition using LOESS)	Wavelet Reconstruction	SSA (Singular Spectrum Analysis)
Category	Decomposition-based	Frequency-based	Subspace-based
Core Idea	Decomposes the signal into trend, seasonal, and residual components.	Performs multi-resolution analysis by decomposing the signal into different frequency components.	Transforms the time series into a trajectory matrix and extracts dominant components .
Preserves	Trend & seasonal patterns	Local & global frequency information	Latent temporal patterns
Main Benefit	Reduces irregular fluctuations and highlights meaningful trend and seasonality.	Reduces high-frequency noise while preserving important frequency patterns.	Captures hidden structures and filters out noise by reconstructing key components.

Why these three methods?

- **Trend decomposition** to capture long-term and seasonal behavior.
- **Frequency decomposition** to reduce noise and retain multi-scale information.
- **Structural decomposition** to uncover dominant latent patterns.

Together, they capture **complementary aspects of temporal information**.

Raw Time Series



STL



Wavelet



SSA


Higher-Quality
Input for ICTSP

Mathematical Intuition Behind Reconstruction

STL

Decomposition Model

$$X_t = T_t + S_t + R_t$$

Trend + Seasonal + Residual

Objective

- Preserve trend (T_t)
- Preserve seasonality (S_t)
- Suppress irregular residual noise (R_t)

Wavelet

Multi-Resolution Analysis

$$X = A_J + \sum_{j=1}^J \cdot D_j$$

Approximation + Detail Components

Objective

- Keep important low-frequency information (Approximation)
- Remove noisy high-frequency details (Details)

SSA

Low-Rank Signal Reconstruction

$$\mathbf{X} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T$$

Singular Value Decomposition

Objective

- Extract dominant temporal structures (Large singular components)
- Remove weak noisy components (Small singular components)

Different
Mathematical Principles



Different
Signal Representations



Higher-Quality
Temporal Tokens



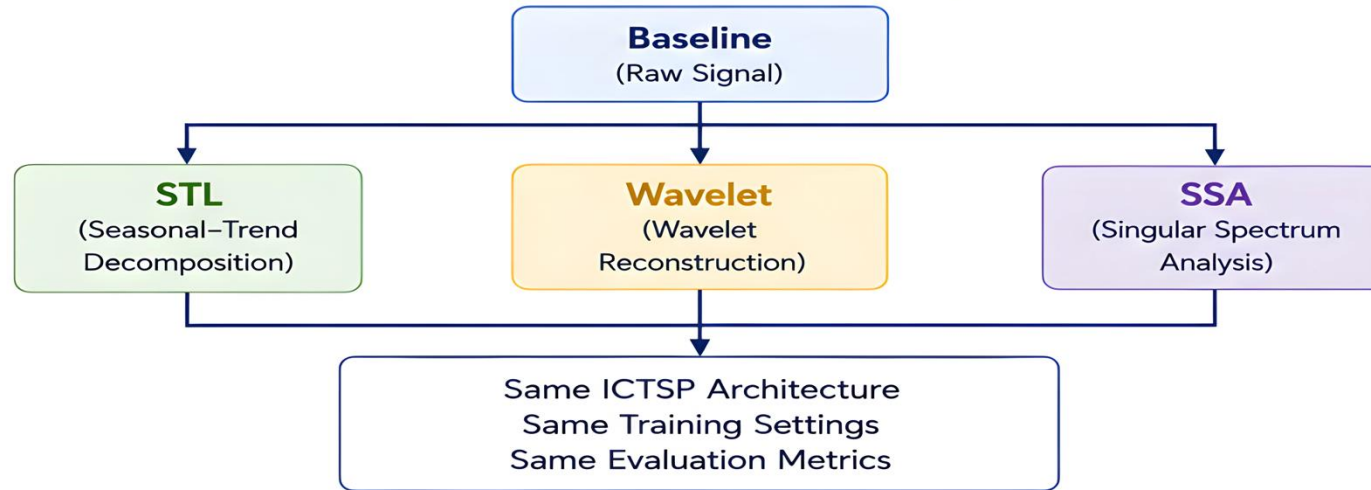
Improved
ICTSP Forecasting

Despite using different mathematical principles, all three methods aim to improve the quality of the temporal representation before ICTSP tokenization.

Experimental Design

A controlled experimental design was adopted to isolate the effect of feature reconstruction.

Controlled Experimental Design



Experimental Regimes

Full Data

Model is trained using the entire training data.

Few-Shot (10% & 5%)

Model is trained with only 10% or 5% of the training data.

Zero-Shot Transfer

Model is directly evaluated on a different dataset without fine-tuning.

Forecast Horizons

96

time steps

192

time steps

Two commonly used long-term forecasting horizons.

Evaluation Metrics

Mean Squared Error (MSE)

Measures the average squared difference between the predicted and true values. Lower values indicate better accuracy.

Mean Absolute Error (MAE)

Measures the average absolute difference between the predicted and true values. Lower values indicate better accuracy.

Datasets

We evaluate the proposed framework on six benchmark datasets from diverse domains and temporal resolutions.

Dataset	Domain	Temporal Resolution	# Variables (Multivariate)	Description
ETTh1	Electricity Transformer	Hourly	7	Electricity load and transformer metrics
ETTh2	Electricity Transformer	Hourly	7	Electricity load and transformer metrics
ETTm1	Electricity Transformer	15 minutes	7	High-frequency electricity monitoring data
ETTm2	Electricity Transformer	15 minutes	7	High-frequency electricity monitoring data
Exchange	Exchange Rates	Daily	8	Foreign exchange rate data
Weather	Weather Measurements	10 minutes	21	Multi-variate weather observations

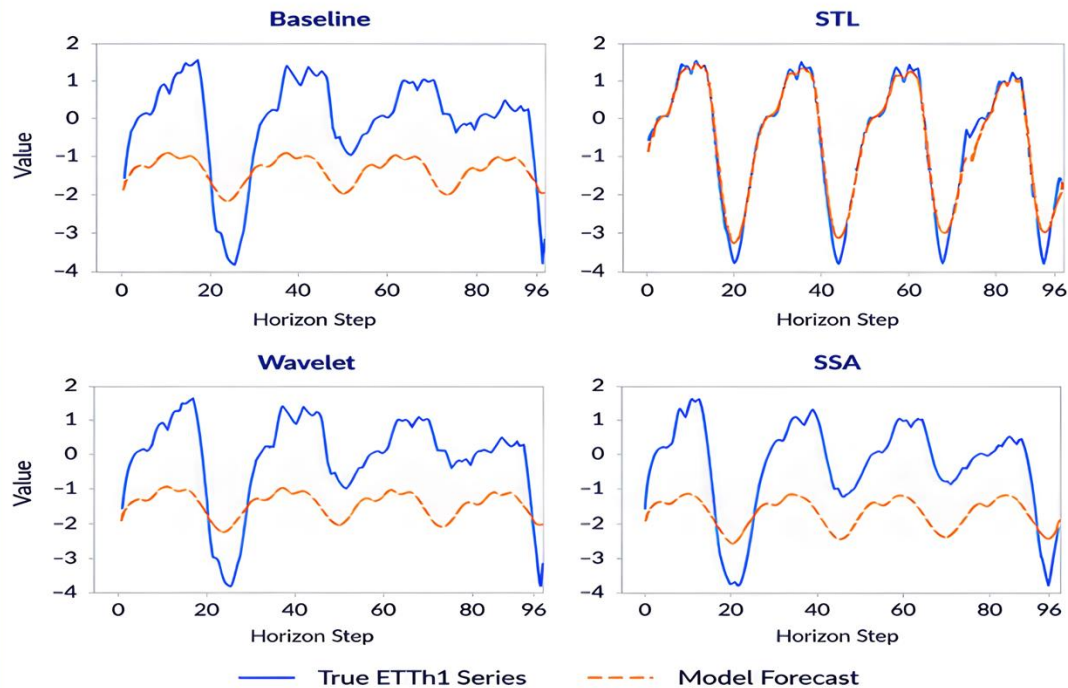


Summary

The datasets span multiple real-world domains such as electricity, finance, and weather at different temporal resolutions from hourly to 10 minutes. This diversity allows us to thoroughly evaluate our framework across a wide range of data characteristics and application scenarios.

Full-Data Results (Prediction Length = 96)

ETTh1 – Forecasting Results (Prediction Length = 96)



The four panels show the **true ETTh1 series** (blue solid line) and the model forecasts (orange dashed line) over a 96-step horizon under full-data setting.

STL reconstruction captures the trend and seasonal patterns most accurately, closely following the true series.

Wavelet provides a good fit, while **Baseline** and **SSA** exhibit larger deviations, particularly at peaks and sharp drops.

Full-Data Forecasting Results (Prediction Length = 96)

Dataset	Baseline		STL		Wavelet		SSA	
	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
ETTh1	0.938057	0.650957	0.173382	0.286589	0.926573	0.643943	0.807237	0.594604
ETTh2	0.205974	0.356015	0.089959	0.195488	0.149553	0.277567	0.202928	0.356269
ETTh1	0.321181	0.382176	0.047325	0.144660	0.326830	0.387153	0.278753	0.345647
ETTh2	0.113306	0.246466	0.017341	0.092325	0.088804	0.209247	0.096893	0.222022
Exchange	0.078114	0.197856	0.079666	0.199849	0.076366	0.195369	0.807428	0.754650
Weather	0.171992	0.226515	0.119284	0.184203	0.166871	0.221704	0.158442	0.214336

Lower values are better. Best results for each dataset are shown in **bold**.

Key Observations

- ✓ **STL** reconstruction achieves the best performance on **ETTh1** and **ETTh2**, producing the lowest MSE and MAE.
- ✓ **Wavelet** reconstruction is competitive on **ETTh2** and **Exchange**, delivering the best results among **all methods** in these datasets.
- ✓ Reconstruction methods generally **improve** forecasting accuracy compared to the **Baseline**, demonstrating the benefit of enhancing signal quality before tokenization.



Full-Data Results (Prediction Length = 192)

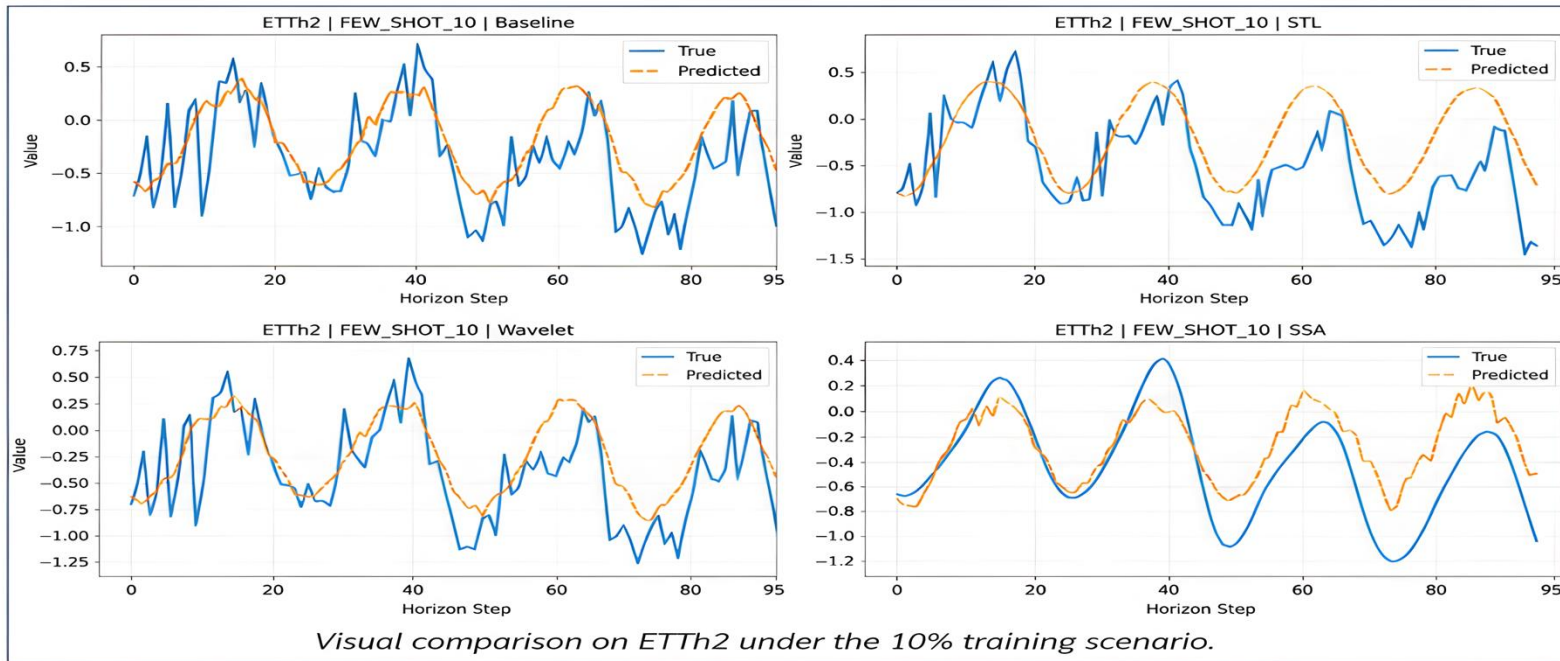
Dataset	Baseline		STL		Wavelet		SSA	
	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
ETTh1	1.216193	0.762107	0.727067	0.583802	1.199919	0.753283	1.092648	0.703179
ETTh2	0.247660	0.357774	0.237452	0.387525	0.216543	0.343457	0.236255	0.346113
ETTM1	0.384293	0.427682	0.102280	0.219533	0.379592	0.419084	0.351916	0.403053
ETTM2	0.134331	0.267424	0.039509	0.136153	0.111222	0.237785	0.118832	0.247476
Exchange	0.159674	0.290677	0.278558	0.407956	0.338611	0.447502	0.178143	0.304663
Weather	0.287441	0.318442	0.201337	0.267551	0.276520	0.309667	0.254118	0.296044

Key Takeaways

- **STL** achieved the best performance on ETTh1, ETTm1, ETTm2, and Weather.
- **Wavelet** was the best on ETTh2, showing strong performance on this dataset.
- **Baseline** ICTSP achieved the lowest error on Exchange dataset.

Few-Shot Learning Results (10% Training Data)

Prediction Horizon = 96



Few-Shot (10%) Summary

Dataset	Best Method
ETTh1	STL
ETTh2	Wavelet
ETTm1	STL
ETTm2	STL
Exchange	Wavelet
Weather	STL

Key Observations

- STL remained the most robust method across the majority of datasets.
- ETTh2 exhibited substantial instability under limited-data conditions.
- Wavelet achieved the best result on ETTh2 and Exchange, showing that reconstruction effectiveness depends on dataset characteristics.

Performance generally degrades under limited supervision, but feature reconstruction continues to improve contextual representations before ICTSP tokenization.

Few-Shot Learning (5% Training Data)

Prediction Horizon = 96

Forecasting Results (Few-Shot 5% Training Data, Prediction Length = 96)									Best Method (Few-Shot 5%)	
Dataset	Baseline		STL		Wavelet		SSA		Dataset	Best Method
	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE		
ETTh1	1.368149	0.806287	1.160687	0.723152	1.370548	0.888004	1.214145	0.754894	ETTh1	STL
ETTh2	25.490588	1.405388	55.838852	1.678296	25.489906	1.405275	94.246677	1.980243	ETTh2	Baseline
ETTm1	2.873104	0.972084	2.281754	0.856409	2.879335	0.970778	2.694124	0.924440	ETTm1	STL
ETTm2	4.487390	0.821114	4.132627	0.688152	4.301528	0.790445	8.103289	0.914773	ETTm2	SSA
Exchange	4.912048	0.768652	4.883196	0.764585	4.872119	0.761928	5.738412	0.842380	Exchange	Baseline/Wavelet
Weather	24.843162	1.229843	3.084274	0.996844	5.746031	1.318005	22.104573	1.187055	Weather	STL

Lower values indicate better performance. The best results for each dataset are shown in **dark blue**.

Key Observations

- **STL** remains the most robust method under extremely limited training data (5%).
- **ETTh2** continues to be highly unstable across all methods.
- Feature reconstruction still improves ICTSP forecasting by strengthening temporal context even with minimal examples.

Zero-Shot Transfer Learning Results

Experimental Settings

Cross-domain

ETTh1 $\longleftrightarrow$ ETTh2

Cross-resolution

ETTh1 $\longleftrightarrow$ ETTm1

Different application domains

Exchange $\longleftrightarrow$ Weather

Main Observations

- **STL achieved the best performance for ETTh1 $\rightarrow$ ETTh2**
 - 96: 0.782 / 0.552
 - 192: 1.314 / 0.734
- **SSA performed best for reverse transfer (ETTh2 $\rightarrow$ ETTh1)**
 - 96: 0.201 / 0.326
 - 192: 0.212 / 0.340
- **Baseline remained strongest for ETTh1 $\rightarrow$ ETTm1**
 - Cross-resolution transfer is more challenging.
- **STL improved Weather $\rightarrow$ Exchange transfer**
 - Better robustness across different domains.

Key Finding

STL dominated most zero-shot experiments, while SSA achieved the best performance in several difficult reverse-transfer tasks.

Key Findings and Contributions

Key Findings

- Feature reconstruction improves ICTSP forecasting performance.
- STL achieved the most consistent performance across multiple datasets.
- Reconstruction remained effective under Full-Data, Few-Shot, and Zero-Shot settings.
- Better signal quality before tokenization improves contextual representations.

Contributions

- Proposed a reconstruction-enhanced ICTSP framework.
- Integrated STL, Wavelet, and SSA without modifying the original ICTSP model.
- Conducted a systematic evaluation on six benchmark datasets.
- Validated the framework under Full-Data, Few-Shot, and Zero-Shot learning scenarios.

Main Contribution

Improving temporal representations before tokenization can enhance ICTSP forecasting performance without changing the original transformer architecture.

Limitations, Future Work and Conclusion

Limitations

- Evaluation was performed using a single forecasting backbone (ICTSP).
- Experiments were limited to forecast horizons of 96 and 192 time steps.
- Reconstruction methods (STL, Wavelet, SSA) used fixed hyperparameters.
- Computational constraints limited evaluation on larger horizons and broader benchmarks.

Future Work

- Extend feature reconstruction to other transformer-based forecasting models.
- Develop adaptive or learnable reconstruction mechanisms.
- Evaluate longer prediction horizons and additional real-world datasets.
- Jointly optimize reconstruction and tokenization within the forecasting pipeline.

Conclusion

- ✓ Feature reconstruction before tokenization **improves ICTSP forecasting performance.**
- ✓ **STL decomposition** delivered the most consistent improvements across Full-Data and Zero-Shot settings.
- ✓ Enhancing signal quality leads to **better contextual token representations** and **more robust forecasting.**

Preprocessing matters:

Improving the input representation can significantly enhance in-context time-series forecasting.



Thank You!

Questions are welcome.

Suggestions and **feedback** are always appreciated.

